

C24 SYNTHETIC SAFETY MANUAL

Revision 1.0

Valid from batch 68

SCANNED COVER SHEET - NO TEXT LAYER

This sheet was photocopied onto the front of the manual.

It carries no text layer, so nothing on it can be verified.

There is no OCR anywhere in this system (Spec §2.2).

General information and symbols

K103 - SI4

General information and symbols

K104 - SI1

General information and symbols

K105 - SI2

General information and symbols

K106 - SI3

General information and symbols

K107 - SI4

General information and symbols

K108 - SI1

Front matter and general information

Machine layout and access points

SF 1.9C) MACHINE LAYOUT AND ACCESS POINTS STAGE 7 ---> Category 3 PL=D Reached

B209 --> K109 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q59 (TELEMECANIQUE LC1-D38BL)

S309 (Emergency stop button in control panel)

EAO 84-5140.1009 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K109 on the machine layout and access points.

Front matter and general information

Machine layout and access points

SF 1.10A) MACHINE LAYOUT AND ACCESS POINTS STAGE 8 ---> Category 3 PL=D Reached

B210 --> K110 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q60 (TELEMECANIQUE LC1-D38BL)

S310 (Emergency stop button in control panel)

EAO 84-5140.1010 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K110 on the machine layout and access points.

Front matter and general information

Machine layout and access points

SF 1.11a) MACHINE LAYOUT AND ACCESS POINTS STAGE 9 ---> Category 3 PL=D Reached

B211 --> K111 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q61 (TELEMECANIQUE LC1-D38BL)

S311 (Emergency stop button in control panel)

EAO 84-5140.1011 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K111 on the machine layout and access points.

Front matter and general information

Machine layout and access points

SF 1.12B) MACHINE LAYOUT AND ACCESS POINTS STAGE 10 ---> Category 3 PL=D Reached

B212 --> K112 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q62 (TELEMECANIQUE LC1-D38BL)

S312 (Emergency stop button in control panel)

EAO 84-5140.1012 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K112 on the machine layout and access points.

Front matter and general information

Emergency stop chain

SF 1.13b) EMERGENCY STOP CHAIN STAGE 11 ---> Category 3 PL=D Reached

B213 --> K113 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q63 (TELEMECANIQUE LC1-D38BL)

S313 (Emergency stop button in control panel)

EAO 84-5140.1013 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K113 on the emergency stop chain.

Front matter and general information

Emergency stop chain

SF 1.14C) EMERGENCY STOP CHAIN STAGE 12 ---> Category 3 PL=D Reached

B214 --> K114 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q64 (TELEMECANIQUE LC1-D38BL)

S314 (Emergency stop button in control panel)

EAO 84-5140.1014 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K114 on the emergency stop chain.

Safety functions of the C24 cell

Emergency stop chain

SF 2.15A) EMERGENCY STOP CHAIN STAGE 13 ---> Category 3 PL=D Reached

B215 --> K115 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q65 (TELEMECANIQUE LC1-D38BL)

S315 (Emergency stop button in control panel)

EAO 84-5140.1015 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K115 on the emergency stop chain.

Safety functions of the C24 cell

Emergency stop chain

SF 2.16a) EMERGENCY STOP CHAIN STAGE 14 ---> Category 3 PL=D Reached

B216 --> K116 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q66 (TELEMECANIQUE LC1-D38BL)

S316 (Emergency stop button in control panel)

EAO 84-5140.1016 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K116 on the emergency stop chain.

Safety functions of the C24 cell

Emergency stop chain

SF 2.17B) EMERGENCY STOP CHAIN STAGE 15 ---> Category 3 PL=D Reached

B217 --> K117 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q67 (TELEMECANIQUE LC1-D38BL)

S317 (Emergency stop button in control panel)

EAO 84-5140.1017 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K117 on the emergency stop chain.

Safety functions of the C24 cell

Guard door interlocks

SF 2.18b) GUARD DOOR INTERLOCKS STAGE 16 ---> Category 3 PL=D Reached

B218 --> K118 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q68 (TELEMECANIQUE LC1-D38BL)

S318 (Emergency stop button in control panel)

EAO 84-5140.1018 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K118 on the guard door interlocks.

Safety functions of the C24 cell

Guard door interlocks

SF 2.19C) GUARD DOOR INTERLOCKS STAGE 17 ---> Category 3 PL=D Reached

B219 --> K119 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q69 (TELEMECANIQUE LC1-D38BL)

S319 (Emergency stop button in control panel)

EAO 84-5140.1019 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K119 on the guard door interlocks.

Safety functions of the C24 cell

Guard door interlocks

SF 2.20A) GUARD DOOR INTERLOCKS STAGE 18 ---> Category 3 PL=D Reached

B220 --> K120 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q70 (TELEMECANIQUE LC1-D38BL)

S320 (Emergency stop button in control panel)

EAO 84-5140.1020 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K120 on the guard door interlocks.

Safety functions of the C24 cell

Guard door interlocks

SF 2.21a) GUARD DOOR INTERLOCKS STAGE 19 ---> Category 3 PL=D Reached

B221 --> K121 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q71 (TELEMECANIQUE LC1-D38BL)

S321 (Emergency stop button in control panel)

EAO 84-5140.1021 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K121 on the guard door interlocks.

Safety functions of the C24 cell

Guard door interlocks

SF 2.22B) GUARD DOOR INTERLOCKS STAGE 20 ---> Category 3 PL=D Reached

B222 --> K122 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q72 (TELEMECANIQUE LC1-D38BL)

S322 (Emergency stop button in control panel)

EAO 84-5140.1022 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K122 on the guard door interlocks.

Safety functions of the C24 cell

Two-hand control and enabling devices

SF 2.23b) TWO-HAND CONTROL AND ENABLING DEVICES STAGE 21 ---> Category 3 PL=D Reached

B223 --> K123 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q73 (TELEMECANIQUE LC1-D38BL)

S323 (Emergency stop button in control panel)

EAO 84-5140.1023 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K123 on the two-hand control and enabling devices.

Safety functions of the C24 cell

Two-hand control and enabling devices

SF 2.24C) TWO-HAND CONTROL AND ENABLING DEVICES STAGE 22 ---> Category 3 PL=D Reached

B224 --> K124 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q74 (TELEMECANIQUE LC1-D38BL)

S324 (Emergency stop button in control panel)

EAO 84-5140.1024 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K124 on the two-hand control and enabling devices.

Safety functions of the C24 cell

Two-hand control and enabling devices

SF 2.25A) TWO-HAND CONTROL AND ENABLING DEVICES STAGE 23 ---> Category 3 PL=D Reached

B225 --> K125 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q75 (TELEMECANIQUE LC1-D38BL)

S325 (Emergency stop button in control panel)

EAO 84-5140.1025 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K125 on the two-hand control and enabling devices.

Safety functions of the C24 cell

Two-hand control and enabling devices

SF 2.26a) TWO-HAND CONTROL AND ENABLING DEVICES STAGE 24 ---> Category 3 PL=D Reached

B226 --> K126 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q76 (TELEMECANIQUE LC1-D38BL)

S326 (Emergency stop button in control panel)

EAO 84-5140.1026 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after

every replacement of K126 on the two-hand control and enabling devices.

Safety functions of the C24 cell

Light curtain muting

SF 2.27B) LIGHT CURTAIN MUTING STAGE 25 ---> Category 3 PL=D Reached

B227 --> K127 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q77 (TELEMECANIQUE LC1-D38BL)

S327 (Emergency stop button in control panel)

EAO 84-5140.1027 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K127 on the light curtain muting.

Safety functions of the C24 cell

Light curtain muting

SF 2.28b) LIGHT CURTAIN MUTING STAGE 26 ---> Category 3 PL=D Reached

B228 --> K128 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q78 (TELEMECANIQUE LC1-D38BL)

S328 (Emergency stop button in control panel)

EAO 84-5140.1028 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K128 on the light curtain muting.

Electrical references and part numbers

Light curtain muting

SF 3.29C) LIGHT CURTAIN MUTING STAGE 27 ---> Category 3 PL=D Reached

B229 --> K129 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q79 (TELEMECANIQUE LC1-D38BL)

S329 (Emergency stop button in control panel)

EAO 84-5140.1029 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K129 on the light curtain muting.

Electrical references and part numbers

Light curtain muting

SF 3.30A) LIGHT CURTAIN MUTING STAGE 28 ---> Category 3 PL=D Reached

B230 --> K130 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q80 (TELEMECANIQUE LC1-D38BL)

S330 (Emergency stop button in control panel)

EAO 84-5140.1030 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K130 on the light curtain muting.

Electrical references and part numbers

Light curtain muting

SF 3.31a) LIGHT CURTAIN MUTING STAGE 29 ---> Category 3 PL=D Reached

B231 --> K131 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q81 (TELEMECANIQUE LC1-D38BL)

S331 (Emergency stop button in control panel)

EAO 84-5140.1031 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K131 on the light curtain muting.

Electrical references and part numbers

Safety relay wiring

SF 3.32B) SAFETY RELAY WIRING STAGE 30 ----> Category 3 PL=D Reached

B232 --> K132 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q82 (TELEMECANIQUE LC1-D38BL)

S332 (Emergency stop button in control panel)

EAO 84-5140.1032 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K132 on the safety relay wiring.

Electrical references and part numbers

Safety relay wiring

SF 3.33b) SAFETY RELAY WIRING STAGE 31 ----> Category 3 PL=D Reached

B233 --> K133 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q83 (TELEMECANIQUE LC1-D38BL)

S333 (Emergency stop button in control panel)

EAO 84-5140.1033 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K133 on the safety relay wiring.

Electrical references and part numbers

Safety relay wiring

SF 3.34C) SAFETY RELAY WIRING STAGE 32 ---> Category 3 PL=D Reached

B234 --> K134 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q84 (TELEMECANIQUE LC1-D38BL)

S334 (Emergency stop button in control panel)

EAO 84-5140.1034 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K134 on the safety relay wiring.

Electrical references and part numbers

Safety relay wiring

SF 3.35A) SAFETY RELAY WIRING STAGE 33 ---> Category 3 PL=D Reached

B235 --> K135 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q85 (TELEMECANIQUE LC1-D38BL)

S335 (Emergency stop button in control panel)

EAO 84-5140.1035 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K135 on the safety relay wiring.

Electrical references and part numbers

Safety relay wiring

SF 3.36a) SAFETY RELAY WIRING STAGE 34 ----> Category 3 PL=D Reached

B236 --> K136 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q86 (TELEMECANIQUE LC1-D38BL)

S336 (Emergency stop button in control panel)

EAO 84-5140.1036 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K136 on the safety relay wiring.

Electrical references and part numbers

Part numbers and suppliers

SF 3.37B) PART NUMBERS AND SUPPLIERS STAGE 35 ---> Category 3 PL=D Reached

B237 --> K137 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q87 (TELEMECANIQUE LC1-D38BL)

S337 (Emergency stop button in control panel)

EAO 84-5140.1037 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K137 on the part numbers and suppliers.

Electrical references and part numbers

Part numbers and suppliers

SF 3.38b) PART NUMBERS AND SUPPLIERS STAGE 36 ---> Category 3 PL=D Reached

B238 --> K138 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q88 (TELEMECANIQUE LC1-D38BL)

S338 (Emergency stop button in control panel)

EAO 84-5140.1038 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K138 on the part numbers and suppliers.

Electrical references and part numbers

Part numbers and suppliers

SF 3.39C) PART NUMBERS AND SUPPLIERS STAGE 37 ---> Category 3 PL=D Reached

B239 --> K139 - SI4

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q89 (TELEMECANIQUE LC1-D38BL)

S339 (Emergency stop button in control panel)

EAO 84-5140.1039 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K139 on the part numbers and suppliers.

Electrical references and part numbers

Part numbers and suppliers

SF 3.40A) PART NUMBERS AND SUPPLIERS STAGE 38 ---> Category 3 PL=D Reached

B240 --> K140 - SI1

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q90 (TELEMECANIQUE LC1-D38BL)

S340 (Emergency stop button in control panel)

EAO 84-5140.1040 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K140 on the part numbers and suppliers.

Electrical references and part numbers

Part numbers and suppliers

SF 3.41a) PART NUMBERS AND SUPPLIERS STAGE 39 ---> Category 3 PL=D Reached

B241 --> K141 - SI2

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q91 (TELEMECANIQUE LC1-D38BL)

S341 (Emergency stop button in control panel)

EAO 84-5140.1041 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K141 on the part numbers and suppliers.

Electrical references and part numbers

Part numbers and suppliers

SF 3.42B) PART NUMBERS AND SUPPLIERS STAGE 40 ---> Category 3 PL=D Reached

B242 --> K142 - SI3

(B&R X20SI4100 - Safe Dig. In.- 2 channel wiring)

Q92 (TELEMECANIQUE LC1-D38BL)

S342 (Emergency stop button in control panel)

EAO 84-5140.1042 SCHNEIDER ZB4-BS844

The stop category is verified at commissioning and after every replacement of K142 on the part numbers and suppliers.